

DATA ANALYTICS CASE STUDY

Global Customer Churn & Retention Analytics using SQL + Power BI

An end-to-end analytics build — from a 1.37M-row relational dataset in SQL Server to a 3-page decision-ready Power BI dashboard — uncovering where, why, and how much churn is costing an international subscription business.

55K

CUSTOMERS

1.37M

FACT ROWS

30

COUNTRIES

\$31.5M

REVENUE TRACKED

1 Project Overview

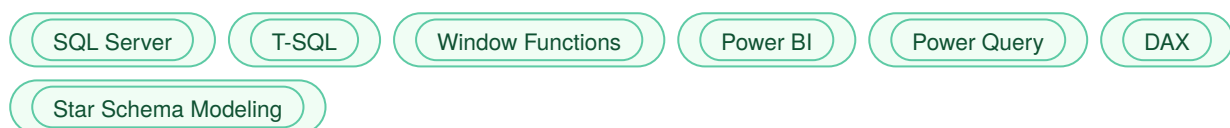
Customer churn is one of the most expensive problems in any subscription-based business — losing a customer doesn't just erase future revenue, it wastes the acquisition cost already spent to win them. This project builds a complete analytics pipeline for an international subscription business, simulating a realistic 55,000-customer base spread across 30 countries, to answer one core question:

Where is churn happening, why is it happening, and how much revenue is genuinely at risk?

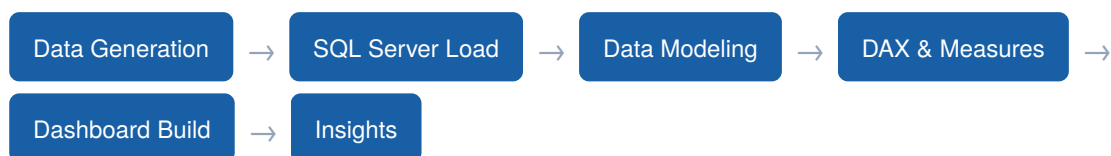
Business Questions Addressed

- What is the overall churn rate, and how does it vary by country, contract type, and subscription plan?
- Which currently active customers carry the highest risk of churning?
- How much monthly revenue is tied to customers in their final (churn) month?
- Which acquisition channels bring in customers who stay the longest and spend the most?
- How does retention trend as customer tenure increases?

Tech Stack



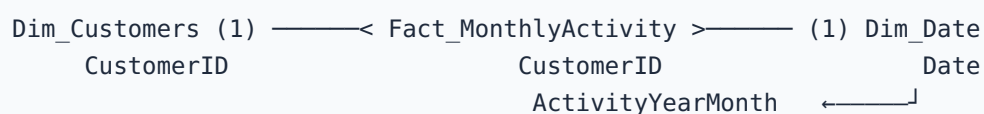
End-to-End Workflow



2 Data Architecture

A star-schema model was designed and loaded into SQL Server, then connected live to Power BI for modeling and visualization.

Table	Rows	Grain
Dim_Customers	55,000	1 row per customer
Fact_MonthlyActivity	1,320,680	1 row per customer per active month
Dim_Date	~1,460	1 row per calendar day



3 Methodology

3.1 SQL Server — Data Engineering

- Designed normalized CREATE TABLE schemas with correct data types for both dimension and fact tables
- Loaded 1.37M+ rows using BULK INSERT with UTF-8 codepage handling
- Wrote validation queries to confirm row counts and referential integrity against the customer dimension
- Built analytical T-SQL queries mirroring every dashboard KPI — including **CTEs**, **window functions** (LAG for month-over-month revenue growth), and conditional aggregation for churn-rate matrices

3.2 Power BI — Modeling

- Connected live to SQL Server in **Import mode** for full DAX/calculated-column flexibility
- Built a dedicated Dim_Date calendar table via DAX for time-intelligence support
- Established a clean star-schema relationship model — single-direction, many-to-one — between fact and dimension tables
- Added an inactive Dim_Date[Date] → Dim_Customers[SignupDate] relationship (activated via USERRELATIONSHIP) to independently analyze signup cohorts

3.3 Calculated Columns

Column	Table	Purpose
TenureBucket	Dim_Customers	Groups customers into 0-6, 7-12, 13-24, 25+ month bands
AgeGroup	Dim_Customers	Segments customers into standard age bands
ChurnStatus	Dim_Customers	Readable Active / Churned label
HighRiskFlag	Dim_Customers	Flags low satisfaction + month-to-month customers
RevenueCategory	Fact_MonthlyActivity	Buckets billed amount into Low / Medium / High

3.4 Key DAX Measures

Measure	Logic
Churn Rate %	Churned Customers ÷ Total Customers
Retention Rate %	1 – Churn Rate %
Revenue at Risk	SUM of billed amount where IsChurnMonth = 1
Revenue MoM Growth %	DATEADD time-intelligence vs. prior month
Customer Lifetime Value	Avg Monthly Charges × Avg Tenure (Months)
New Customers This Month	USERRELATIONSHIP-activated signup-date count

4 Dashboard Design

The final deliverable is a 3-page interactive Power BI dashboard, each page built for a distinct audience and decision.

Page 1 — Executive Overview

Headline KPIs (customers, churn rate, retention rate, revenue, NPS), monthly revenue & active-customer trend, churn-reason ranking, revenue by plan, churn rate by contract type, and a regional slicer panel.

Page 2 — Churn Intelligence

A high-risk customer watchlist, satisfaction-vs-NPS distribution, ranked churn drivers, a contract-type × plan-type churn heatmap, and a tenure-based retention curve.

Page 3 — Customer Value Analysis

Revenue trend with month-over-month growth, customer lifetime value, revenue-at-risk, acquisition-channel performance, and contract-level risk exposure.

5 Key Outcomes

26.05%

OVERALL CHURN RATE

73.95%

RETENTION RATE

\$31.5M

REVENUE TRACKED

\$344.6K

REVENUE AT RISK

- **Contract type is the single strongest churn predictor:** month-to-month customers churn at **38.6%**, versus just **6.3%** for two-year contracts — a clear signal to prioritize contract-upgrade incentives.
- **Price and competition dominate churn reasons:** "Price too high" and "better competitor offer" together account for over **5,200 lost customers**, pointing to a pricing/positioning review as the highest-leverage retention lever.
- **A repeatable High-Risk Customer view** (low satisfaction + month-to-month contract) enables a retention team to run proactive, targeted outreach rather than reactive win-back campaigns.
- **Revenue concentration by plan** shows Premium and Standard tiers driving the majority of tracked revenue, informing where retention investment pays back fastest.
- **Retention decays sharply in the first 12 months** before stabilizing — the tenure curve highlights the first year as the critical window for engagement efforts.

6 Skills Demonstrated

SQL Server

- Schema design & data typing
- Bulk data loading
- CTEs & window functions
- Analytical query writing

Power BI

- Star-schema data modeling
- DAX measures & calculated columns
- Time intelligence
- Dashboard & UX design